cargo delivery missions, demonstrating the potential of UAV-based logistics.

- *Public safety and disaster response*: Police and emergency responders utilized UAVs for crowd monitoring, fire detection, and hazardous environment assessments, enhancing operational safety and efficiency.

4.4.4 Communication and technological applications

The integration of UAVs within broader communication and data networks highlighted their role as mobile aerial nodes in emerging global information systems.

- *Data relay and network extension*: Drones served as communication relays in remote or disaster-stricken areas, providing emergency data connectivity where terrestrial infrastructure was unavailable.
- *Technology demonstration platforms*: Third-generation UAVs acted as testbeds for AI algorithms, autonomous flight systems, and sensor integration experiments, accelerating advancements for future UAV generations.
- *Satellite communication testing*: Equipped with SATCOM antennas, drones were used to validate satellite link reliability and test next-generation data relay technologies for aerospace and defense sectors.
- *Cybersecurity and encryption research*: UAVs contributed to developing and testing secure data transmission protocols and anti-jamming communication techniques, ensuring reliability in contested electronic environments.